

Extending Theorem 2 of *Robust Trust* to infinite M and Θ : A Pareto-frontier addendum to the menu engine (v9, Pass 3)

Pedro Aldighieri

21 May 2026

Abstract

This note is the v9 layer above the v8 menu-engine exposition. It records the third pass at the infinite extension of Theorem 2 of Dworczak & Smolin’s *Robust Trust* (2026), motivated by Piotr Dworczak’s 2026-05-20 suggestion to reformulate the agent’s strategy as a compact subset of the weak Pareto frontier W^P of state-contingent payoff profiles, with the misaligned adviser mapping signals directly into payoff vectors in the chosen subset. We report four new results. **(T1) Finite-menu Pareto-Hall calibration** closes *unconditionally* in payoff-label coordinates: at any finite-menu Pareto-completed ambient local maximizer of the value functional, Clarke–Danskin stationarity produces measurable active-face weights $\lambda^\pm : M \rightarrow \Delta(k)$ whose normalized images $p_i = g_i/q_i$ lie in the Bayes-cone $B_W(w_i)$. Calibration *emerges* as a Lagrange multiplier rather than being imposed as an external Hall-type assumption. **(T2) $\alpha = 0$ unconditional** infinite- M Theorem 2 closes via a singleton-strategy reduction: when the adviser is purely adversarial, the optimal payoff-profile menu collapses to a single Bayes-optimal-at-prior profile, and the adviser’s fixed-message strategy makes the agent’s continuation Bayes-optimal at the prior. **(T3) Compact-menu Pareto-Hall** closes in payoff-label coordinates under finite effective exposure of the optimal menu, a primitive structural condition strictly weaker than v8 menu-Hall in those coordinates. **(T4) Honest endpoint.** For general $\alpha \in (0, 1)$ and the original-game Definition 2, the lift from payoff-label calibration to messagewise calibration requires a finite-fiber matching condition (*D2*) which is shown to be the finite-fiber version of v8’s menu-Hall, not a strictly weaker primitive. The deletion-compatible Hall duality theorem named in the project closure memo remains the single open object. A fresh seven-candidate primitive search (atomless plus fiber-richness, single-valued Gauss map, τ -symmetry, smooth plus strictly convex W^P , product utility, coarsening, fiber-rich Lagrangian transport) did not produce a strictly weaker primitive. All claims above were verified by independent reviewer passes on fresh chats; the pipeline trace is in §??.

1 Setting and recap of v8

We use the notation of *Robust Trust* and of v8 (§?? in `exposition.pdf`). Ω is finite with full-support prior μ_0 ; $s \in \Delta(\Omega)$ has unconditional law τ on $M = \text{supp}(\tau)$; $\theta \in \Theta$ (compact metric) is drawn from $f(\cdot \mid \omega)$, independent of s given ω ; A is compact metric; $u(a, \omega, \theta)$ is bounded continuous in a . With probability $\alpha \in [0, 1]$ the adviser is aligned and reports s truthfully; with probability $1 - \alpha$ misaligned, using any $\beta : M \rightarrow \Delta(M)$ Borel. Agent strategies are $\sigma : M \times \Theta \rightarrow \Delta(A)$ Borel. The robust value is $U^* = \sup_{\sigma \in \Sigma} U(\sigma)$ with U as in v8 Eq. (1).

The v8 menu engine works on the payoff-profile polytope $W = \{w \in \mathbb{R}^N : \exists \hat{\sigma} : \Theta \rightarrow \Delta(A), w(\omega) = \mathbb{E}_{\hat{\sigma}}[u(a, \omega, \theta) \mid \omega]\}$, which is convex compact (*Robust Trust* Lemma 2 of Theorem 1’s proof, p. 27). Its weak Pareto frontier W^P is the closed subset of profiles not strictly state-wise

dominated. Lemma 2 records the key equivalence $\hat{\sigma}$ is Bayes-optimal for some belief iff its profile is in W^P .

$v8$ delivers a three-tier result:

- **Tier 1a (value optimality):** $\sup_C F(C)$ is attained on $\mathcal{K}(W)$ under standing alone, where $F(C) = \int [\alpha \max_C s \cdot w + (1 - \alpha) \min_C s \cdot w] d\tau$.
- **Tier 1b (exact adversary):** under *exact-contact*, a deterministic message kernel $\beta^*(\cdot | s) = \delta_{m^*(s)}$ attains the misaligned infimum.
- **Tier 2 (full robust rationalizability):** under *exact-contact* plus *menu-Hall*, Definition 2 holds q -a.e.

The $v8$ closure memo names the single open object as a *deletion-compatible Hall duality theorem* reconciling sourcewise deletion certificates with messagewise Bayes-cone calibration; both prior attack routes (constrained-persuasion duality and calibration-defect bound) stalled at the same structural family of gaps.

2 The Pareto-frontier reformulation

Piotr’s proposal (2026-05-20, verbatim). “Instead of thinking of the Agent as selecting a mapping from private signals and messages of the Adviser to distributions over actions, we can think of the Agent as selecting a subset of the weak Pareto frontier of state-contingent payoffs, relying on Lemma 2 in the proof of Theorem 1. With some appropriate topology, the set of all feasible subsets is compact. The strategy of the misaligned Adviser is to map their signal into a state-contingent payoff vector in the set chosen by the Agent.”

The game $\mathcal{G}_P$.

- Agent’s strategy space: $\mathcal{K}(W^P) = \{C \subseteq W^P : C \neq \emptyset, C \text{ compact}\}$, equipped with Hausdorff distance. Blaschke selection yields $(\mathcal{K}(W^P), d_H)$ compact metric.
- Misaligned adviser’s strategy space for fixed C : Borel maps $\beta : M \rightarrow W^P$ with $\beta(s) \in C$ τ -a.e. Compactness follows from the relaxed-Young embedding into $\Delta(M \times C)$.
- Payoff $U_P(C, \beta) = \alpha \int_M \max_{w \in C} s \cdot w d\tau + (1 - \alpha) \int_M s \cdot \beta(s) d\tau$.
- Agent’s robust value $V_P(C) = \inf_{\beta} U_P(C, \beta) = \alpha \int \max_C s \cdot w d\tau + (1 - \alpha) \int \min_C s \cdot w d\tau$.

WLOG equivalence to the original value. By Lemma 2 of Theorem 1’s proof, every optimal $\sigma \in \Sigma$ is equivalent to one whose message-conditional profile $w_\sigma(m)$ takes values in W^P for every $m \in M$. Setting $C_\sigma := \overline{w_\sigma(M)} \subseteq W^P$ collapses the adversary’s effective action set to C_σ and yields

$$U^* = \sup_{C \in \mathcal{K}(W^P)} V_P(C). \quad (1)$$

The lift from any $C \in \mathcal{K}(W^P)$ to a measurable original-game strategy is the profile-realization right inverse $R : W \rightarrow \hat{\Sigma}$ (KRN; $v8$ §3 sub-lemma).

What $\mathcal{G}_P$ buys cleanly. The adversary's best-response in $\mathcal{G}_P$ is $\beta^*(s) \in \arg \min_{w \in C} s \cdot w$, existence automatic by Kuratowski–Ryll–Nardzewski on the compact-valued correspondence $s \mapsto C$. This dispenses with the *v8/v5 Asc-lsc* hypothesis. Tier 1a in payoff-vector form is therefore strictly cleaner than *v8*'s.

3 Finite-menu Pareto–Hall calibration via Clarke–Danskin

The key technical novelty of *v9* is a Lagrange-multiplier-based calibration theorem that does not assume any Hall-type kernel existence.

Theorem 1 (Finite-menu Pareto–Hall calibration). *Let $C^* = \{w_1, \dots, w_k\} \subseteq W^P$ be a Pareto-completed ambient local maximizer of the finite-menu functional*

$$F_k(w_1, \dots, w_k) = \int_M [\alpha \max_i s \cdot w_i + (1 - \alpha) \min_i s \cdot w_i] \tau(ds).$$

Then there exist Borel measurable active-face weights

$$\lambda^+ : M \rightarrow \Delta(k), \quad \lambda^- : M \rightarrow \Delta(k),$$

with $\text{supp } \lambda^+(s) \subseteq \arg \max_i s \cdot w_i$ and $\text{supp } \lambda^-(s) \subseteq \arg \min_i s \cdot w_i$ for τ -a.e. s , such that the integral subgradient component

$$g_i = \alpha \int \lambda_i^+(s) s d\tau + (1 - \alpha) \int \lambda_i^-(s) s d\tau$$

satisfies $g_i \in N_W(w_i)$ for every i ; and with $q_i = \alpha \int \lambda_i^+ d\tau + (1 - \alpha) \int \lambda_i^- d\tau$, the normalized vector $p_i := g_i/q_i$ lies in $B_W(w_i) = N_W(w_i) \cap \Delta(\Omega)$ for every i with $q_i > 0$.

The proof chain is twelve lemmas long; here we record the load-bearing structure.

Lemma 2 (Integral Clarke–Danskin representation). *Every $g \in \partial_C F_k(\bar{w})$ admits Borel multipliers $\lambda^\pm : M \rightarrow \Delta(k)$ with $\text{supp } \lambda^\pm(s)$ in the active max/min faces and the integral representation above.*

Proof outline. Apply Clarke's integral subdifferential interchange (Clarke 1983, §2.7) to obtain $\partial_C F_k(\bar{w}) \subseteq \int_M \partial_C \phi_s(\bar{w}) d\tau$ (Aumann integral closed in finite dimension; uniform Lipschitz $L(s) = \|s\|_\infty \leq 1$). Use the larger active-weight correspondence $R(s) \supseteq \partial_C \phi_s(\bar{w})$ rather than equality (*patched after reviewer pass 01*: equality fails on positive-measure tie sets in general). Select a Borel measurable selector $\xi(s) \in R(s)$ on the Borel partition $\{E_{I,J}\}$ of M into active-cell strata, apply Kuratowski–Ryll–Nardzewski to decompose $\xi(s)$ into $\lambda^+(s) \in \Delta(I)$ and $\lambda^-(s) \in \Delta(J)$.

Lemma 3 (Clarke–Fermat normal-cone stationarity). *If $\bar{w}$ is an ambient local maximizer of F_k on W^k , then there exists $g \in \partial_C F_k(\bar{w})$ with $g \in N_{W^k}(\bar{w}) = \prod_i N_W(w_i)$.*

Proof outline. Apply Clarke's necessary condition for the constrained minimum of $-F_k$ on W^k , yielding $0 \in \partial_C (-F_k)(\bar{w}) + N_{W^k}(\bar{w})$, and use the product normal-cone formula. The ambient (rather than frontier-only) local-max hypothesis is essential for N_W -normality and is delivered by Lemma 2 of Theorem 1's proof under standard Pareto-completion.

Lemma 4 (Multipliers are the calibration kernel). *With g from Lemma ?? and $\lambda^\pm$ from Lemma ??, the mass balance $\sum_i q_i = 1$ holds, $p_i := g_i/q_i \in \Delta(\Omega)$ for $q_i > 0$, and $p_i \in B_W(w_i)$.*

Proof outline. *Mass balance:* integrate $\sum_i \lambda_i^\pm = 1$ against τ . *Posterior in simplex:* $\sum_\omega [g_i]_\omega = \alpha \int \lambda_i^+ \sum_\omega s(\omega) d\tau + (1 - \alpha) \int \lambda_i^- \sum_\omega s(\omega) d\tau = q_i$ since $s \in \Delta(\Omega)$; nonneg entries by $s \geq 0$ and $\lambda^\pm \geq 0$. *Calibration:* $g_i \in N_W(w_i)$ by Lemma ??; $N_W(w_i)$ is a convex cone; positive rescaling $p_i = g_i/q_i$ preserves cone membership; intersection with $\Delta(\Omega)$ gives $B_W(w_i)$.

Mechanism. The Clarke multipliers $\lambda^\pm$ are not bookkeeping: they encode *both* the aligned tie-routing weights on the supporting hyperplane $\arg \max_i s \cdot w_i$ and the adversarial tie-routing weights on $\arg \min_i s \cdot w_i$. The normalization $p_i = g_i/q_i$ is the disintegration posterior at label i under a (label-coordinate) joint law combining aligned λ^+ -routing with weight α and adversarial λ^- -routing with weight $1 - \alpha$. Fermat normality of g_i in $N_W(w_i)$ is therefore *equivalent* to Bayes-optimality of the disintegrated posterior at w_i : calibration emerges from the finite-menu optimization itself, not from an external Hall assumption.

No hidden hypotheses. The proof uses no atomlessness of τ , no genericity (positive-measure ties are handled by measurable selection on Borel active-cell strata), no strict convexity of W , and no τ -AC restriction. Reviewer 02 (fresh chat) verified Lemmas ??–?? with all hypotheses on the table.

v8 sharpness package compatibility. Theorem ?? does not contradict v8 Lemma 7 (cone intersection) or Theorem 8 (no-free-dust) on the WTA ternary witness, because the witness is *not* a finite ambient local maximizer of F_k in the Lemma ?? sense. The v8 closure memo classifies the witness as a menu-engine artefact; this classification is what allows Theorem ?? to hold without contradicting the witness’s non-calibration.

4 $\alpha = 0$ unconditional via singleton strategy

When the adviser is purely adversarial, the optimal payoff-profile menu collapses to a single Bayes-optimal-at-prior profile, and Definition 2 closes by a brute-force calculation.

Theorem 5 ($\alpha = 0$ unconditional Theorem 2). *Under the standing hypotheses and $\alpha = 0$ (pure adversarial adviser), there exists a robustly rationalizable optimal strategy without any finiteness of M or Θ . Explicitly:*

- Pick $w_0 \in \arg \max_{w \in W} \mu_0 \cdot w$, attained because W is compact in $\mathbb{R}^N$.
- Set $\hat{\sigma}^*(m) := R(w_0)$ for every $m \in M$ (the agent ignores advice and plays the Bayes-optimal-at-prior continuation).
- Pick any fixed $m_0 \in M$ and set $\beta^*(\cdot \mid s) := \delta_{m_0}$ for every s (the adversary sends a constant message).

Then β^* is adversarial against σ^* (agent’s payoff is β -independent since $\hat{\sigma}^*$ is constant in m), the unique q -positive message is m_0 , and the disintegration posterior there is μ_0 (the prior). By construction $w_0 \in \arg \max_W \mu_0 \cdot w$, hence $R(w_0)$ is Bayes-optimal at μ_0 . Definition 2 holds.

Reviewer 03 (fresh chat) verified Theorem ?? and flagged it as the cleanest infinite extension in the pure adversarial regime. The result strictly generalizes the paper’s Theorem 2 for $\alpha = 0$.

Caveat on scope. $\alpha = 0$ is the *degenerate* limit of the model in which the adviser is never aligned. The substantive regime is $\alpha \in (0, 1)$; Theorem ?? should be read as the easy end of the spectrum, not as the headline contribution.

5 Compact-menu Pareto-Hall under finite effective exposure

The finite-menu Theorem ?? extends to compact menus under an economically meaningful primitive condition strictly weaker than *v8*’s menu-Hall in payoff-label coordinates.

Assumption A1 ((R1) Stratification). W^P admits a finite Borel stratification $W^P = \bigsqcup_{\ell} S_{\ell}$ such that on each stratum’s relative interior the Gauss map $\nu : S_{\ell} \rightarrow S^{N-1}$ (selecting a unit-norm supporting direction) is single-valued and continuous.

Polyhedral W satisfies (R1) trivially with strata = relative interiors of faces. Smooth strictly convex W satisfies (R1) with a single stratum.

Assumption A2 ((R2-FES) Finite effective exposure). At a global maximizer $C^* \in \arg \max_{\mathcal{K}(W^P)} V_P$, there is a finite subset $C_0 = \{w_1, \dots, w_k\} \subseteq C^*$ such that for τ -a.e. $s \in M$, $\max_{w \in C^*} s \cdot w = \max_{w \in C_0} s \cdot w$ and $\min_{w \in C^*} s \cdot w = \min_{w \in C_0} s \cdot w$.

(R2-FES) is the structural translation of “the agent’s effective response is finite-menu, even when the abstract menu is a continuum”. It is economically meaningful in applications with finite-thresholding decision rules (clinical triage, lending tiers).

Theorem 6 (Compact-menu Pareto-Hall). *Under (R1) plus (R2-FES), every global maximizer $C^* \in \arg \max_{\mathcal{K}(W^P)} V_P$ admits Borel multipliers $\lambda^{\pm} : M \rightarrow \Delta(k)$ supported on the finite effective exposure C_0 such that the induced label posteriors $p_i = g_i/q_i$ lie in $B_W(w_i) \cap \Delta(\Omega)$ for every i with $q_i > 0$.*

Proof outline. (R2-FES) gives $V_P(C_0) = V_P(C^*)$; (R1) combined with $C_0 \subset C^*$ makes C_0 a Pareto-completed ambient local maximizer of F_k on the appropriate stratum product; Theorem ?? applies. *Capstone 1* in *v9*’s breakdown 02 (see `breakdown_02_response.md`).

General compact (*v9* Capstone 2). Under a weaker condition (*R3-FCA/PK*) requiring approximation of C^* by finite critical menus in Painlevé–Kuratowski convergence, Capstone 2 lifts the finite-effective theorem to general compact menus. This is conditional; not unconditional.

6 The honest endpoint: $(D2) \equiv$ menu-Hall for general α

The lift from payoff-label calibration (Theorems ?? and ??) to original-game messagewise calibration (Definition 2 in the q -a.e. infinite-space reading) requires a finite-fiber matching condition we call $(D2)$.

Assumption A3 ((D2) Finite-fiber calibrated matching). There exists a Borel original-game adversarial kernel $\hat{\beta}^* : M \rightarrow \Delta(M)$ such that, with $\gamma_{\alpha} := \alpha(\text{id}, \text{id})_{\#} \tau + (1 - \alpha)\tau \otimes \hat{\beta}^*$ and $q := (\gamma_{\alpha})_2$:

(D2.1) *First-marginal match*: for τ -a.e. s , $\hat{\beta}^*(F_i | s) = \lambda_i^-(s)$ where $F_i = (w^*)^{-1}(\{w_i\})$.

(D2.2) *Rowwise-minimizer support*: $\hat{\beta}^*(F_i | s) > 0$ implies $w_i \in \arg \min_{w \in C^*} s \cdot w$.

(D2.3) *Messagewise calibration*: $P_{\hat{\beta}^*}(\cdot | m) \in B_W(w^*(m))$ for q -a.e. $m \in M$.

Proposition 7. *For general $\alpha \in (0, 1)$, (D2) is structurally the finite-fiber version of $v8$'s menu-Hall. It is not a strictly weaker primitive condition.*

Proof outline (verified by reviewer 03). With λ^- fixed, (D2.3) restated in original-game variables matches the $v8$ menu-Hall condition for a kernel supported on $G(s)$ with posterior in $B(m)$ q -a.e. The finite-label coarsening makes the bookkeeping smaller (only k Bayes cones to satisfy), but the underlying source-to-message splitting inside each fiber F_i in the presence of aligned-diagonal mass is the same Hall transport problem. The $v8$ closure memo names this exact object as deletion-compatible Hall duality.

The aligned/misaligned mismatch. For $\alpha > 0$, $p_i = g_i/q_i$ averages aligned tie-routing λ^+ and misaligned λ^- . The original-game messagewise posterior at any representative $m_i \in F_i$ involves only the aligned-conditional-given- m_i distribution (the truthful aligned mass at the literal message $m = s$, not at the label-representative) plus misaligned λ^- . Forcing these to agree messagewise is (D2.3).

Searcher 02's primitive scan. A fresh seven-candidate search (`searcher_02_response.md`) examined:

- (C1) Atomless τ plus fiber-richness ($\tau(F_i) > 0$): too weak.
- (C2) Single-valued Gauss map at C^* : clarifies equations but does not solve them.
- (C3) Diagonal τ -symmetry: a positive island only for one-dimensional or antipodal cases.
- (C4) Smooth plus strictly convex W^P : meaningful regularity, but not sufficient.
- (C5) Product/separable utility: not a useful sufficient condition.
- (C6) Coarsening / pure-aligned-redirect: helps for exact-contact, not for (D2).
- (C7) Fiber-rich Lyapunov plus Lagrangian transport: closest mathematically, but as stated insufficient; when repaired, it becomes (D2)/menu-Hall.

Verdict: *no candidate yields a genuinely primitive sufficient condition for (D2) strictly weaker than menu-Hall, in payoff-label coordinates.*

7 Pipeline trace and verification

All claims in this note were generated and verified via the MathPipeProver smart-scaffolding pipeline, with Extended Pro as the external proof engine. Each prover output was reviewed on a fresh independent chat. The trace:

Pass	Chat ID	Verdict / output
Formalizer 01	6a0e88b1	Reformulation precise; (D2)≡menu-Hall flagged
Literature 01	6a0e8b5e	BUILD; closest tool: Dworczak–Kolotilin
Searcher 01	6a0e8f51	Primary route: Clarke–Danskin stationarity
Breakdown 01	6a0e95c1	12-lemma chain
Prover 01 (L6)	6a0e9994	Integral Clarke–Danskin representation
Reviewer 01 (L6)	6a0e9cfe	PATCH_SMALL — patched with $R(s)$
Prover 02 (L7+L8)	6a0ea088	Fermat + multiplier hinge
Reviewer 02 (L7+L8)	6a0ea3ca	PASS (Theorem ??)
Prover 03 (L12)	6a0ea7b0	$\alpha = 0$ unconditional; $\alpha > 0$ needs (D2)
Breakdown 02 (compact)	6a0ea80b	(R1)+(R2-FES); (R3-FCA/PK)
Reviewer 03 (singleton)	6a0eaf03	PASS (Theorems ??–??)
Searcher 02 (primitives)	6a0eaf16	No strictly weaker (D2) primitive

8 Open problems and recommendations

The single open object remains the deletion-compatible Hall duality theorem. Both the $v8$ closure memo and this $v9$ confirm this from independent attack directions. The named obstructions are Borel-to-compact non-monotonicity, label-fiber lift, and slack discipline in curved W .

Special primitive islands worth attacking. Per searcher 02:

- Binary state ($|\Omega| = 2$): paper Appendix A.6 covers finite M . Push to infinite M , Θ using Theorem ?? plus an order/monotone structure on $\Delta(\Omega)$.
- Antipodal/radial τ -symmetry: spherical case (paper Appendix A.10) plus rotational invariance forces calibration by symmetry-averaging.
- Polyhedral W with finite-faced optimal trust region: $(R1) +$ finite-faceted polyhedrality reduces (R2-FES) to the finite-vertex case; Theorem ?? closes.

Methodological contribution. The Clarke–Danskin finite-menu calibration mechanism (Theorem ??) is portable beyond robust trust to other zero-sum persuasion-like games with finite-effective payoff-profile menus. Calibration as a Lagrange multiplier is, to our knowledge, a new tool in this literature.

What $v9$ deliberately does not claim. Unrestricted infinite Theorem 2 for general $\alpha \in (0, 1)$ without (D2)/menu-Hall. The route’s negative finding is just as load-bearing as the positive ones: *the locked gate is structural across the menu-engine and Pareto-frontier reformulations.*

Acknowledgments

The Pareto-frontier reformulation is due to Piotr Dworczak (private correspondence, 2026-05-20). The Clarke–Danskin calibration mechanism was identified by the searcher pass on 2026-05-21 and verified across the prover/reviewer chain on the same day. All pipeline artifacts are in `03_runs/piotr_pareto_frontier/`.